

UPMANYU JHA

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WORK EXPERIENCE

Quantiphi Analytics Solutions Private Limited (Google Cloud Partner)

Mumbai, IN

Machine Learning Engineer (LLMs, NLP, Computer Vision, Agents & RAG)

Jul 2023 - Present

- Refactored document processing pipeline into scalable microservices, handling 80,000+ documents quarterly, improving classification accuracy by 19.5% through targeted model retrofitting and pipeline optimizations to reduce manual review time by 80%
- Orchestrated** delivery for squads of 6+; standardized model deployment across 5+ business units via reusable custom artifacts, reducing ramp-up time by 20% and improving team delivery scores by 15%
- Delivered 15+ generative modeling pipelines on Google Cloud, enabling scalable automated processing of 1-2K documents daily
 - Reduced model hallucinations by ~85% through domain-specific fine-tuning and cut GCP costs by 15% via resource optimization, ensuring reliability in unstructured document processing
 - Gained 92% F1-score on structured data extraction by implementing few-shot prompting, reducing manual processing time by over 40%
- Architected** 3-month technical roadmaps for 10+ major projects, resolving scalability and reusability bottlenecks to accelerate project delivery 2-3 weeks ahead of scheduled AI/ML lifecycle
- Engineered a scalable Anomaly Detection system for financial transactions, increasing recall by 35% using statistical profiling for signature detection models flagging data integrity risks, cutting manual review queue by 50% while maintaining precision
 - Designed a novel Retrieval-Augmented Generation (RAG) pipeline with fine-tuned LLMs, elevating the F1-score for financial document extraction from 0.55 baseline to 0.90 (+0.35)
 - Created scalable ML infrastructure (MLOps) and CI/CD pipelines via pytest, automating code reviews to increase unit test coverage to 95%, cutting production bugs by 80% and decreasing deployment failures by 50%

Bugcrowd | Hackerone

Mumbai, IN

Security Researcher | Bug Bounty Hunter (Web Application, API)

Jun 2021 - Jul 2023

- Performed Root Cause Analysis (RCA) on 30+ architectural vulnerabilities, scripting automated checks to ensure privacy compliance
- Crafted Python/Docker automation tools, boosting manual testing efficiency by 60% by rapidly identifying architectural security gaps

TECHNICAL SKILLS

- Programming Languages:** Python, Go, SQL (PostgreSQL), GraphQL, C++, C, Git, Bash Script, **Tools:** Github, Flask, Django, Postman, Jira, Rally
- Machine Learning:** PyTorch, TensorFlow, Keras, TensorRT, OpenCV, Scikit-Learn, **Data Visualization:** Matplotlib, Seaborn, Plotly, Pandas, NumPy
- Generative AI:** OpenAI API, Gemini, Pinecone, **Fine-tuning** (LoRA), **Agents** (n8n, LangChain, crewAI), **Hugging Face** (Transformers: BERT, LSTM)
- Cloud & MLOps:** GCP (Doc AI, ADK, Vertex AI, GKE, Storage, BigQuery), **Amazon Web Services:** (SageMaker, S3, EC2, RDS), Docker, Kubernetes

PROJECTS AND RESEARCH

Smart Traffic Management System (Research Prototype) | GANs, Reinforcement Learning (RL) and Computer Vision (CV)

Dec 2025 - Present

- Targeting 10% reduction in average commute time by designing an AI-agent traffic control prototype integrating Object Detection (YOLO) and multi-agent Reinforcement Learning (RL) in SUMO simulation

AstraShield - Disaster Relief Verification System | Python, FastAPI, Gemini Vision AI, React, GCP

Oct 2025 - Nov 2025

- Engineered Forensic Agentic AI achieving a 90% detection rate via Human-in-the-Loop workflow across 100+ real-world samples, utilizing Gemini Vision Pro and Tesseract OCR for multimodal analysis
- Architected real-time pipeline integrating Google Custom Search and WHOIS, delivering background checks and risk scores in under 30 seconds
- Built scalable FastAPI backend implementing Model Context Protocol (MCP) for Agentic function calling with evidence logging; deployed with React frontend for safe donation verification

AlphaFold - Drug Discovery | DeepMind AlphaFold

Jul 2023 - Jan 2024

- Think-Tank (H1-24) Award:** Leveraged AlphaFold research for strategic POC, advancing protein and peptide structure predictions for a pharmaceutical client's Phase-1 trial

LEADERSHIP AND OPEN SOURCE CONTRIBUTION

Teaching Assistant | Mentor - HAWK-i Labs

Aug 2021 - Present

- Co-led Python / C++ / NLP workshops for 1000+ students, significantly improving preparedness for 12th Board and University exams

Project - iSee: Drafted Lightweight, privacy-first Apple utility shared on Swift subreddit; achieved 17.4% engagement among 46k+ viewers

Blogging: Authored technical articles on Medium (hackergod00001), gaining 1,000+ global followers and establishing authority with an audience of 20,000+

EDUCATION

University Of Mumbai

Mumbai, IN

Bachelor of Engineering in Computer Engineering, **Cumulative GPA: 8.92/10** | **Relevant Coursework:** Deep Learning, Computer Vision, NLP, AWS

CERTIFICATIONS AND TRAINING

Google Cloud Professional Cloud Security Engineer (GCP PCSE)

Dec 2025

Google Generative AI Leader (GCP Gen AI)

Dec 2025

Google Cloud Professional Cloud Architect (GCP PCA)

Jun 2025

Google Cloud Professional Machine Learning Engineer (GCP PMLE)

May 2024